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Operating System

OPTIMAL

Aim:

To write a c program to implement Optimal page replacement algorithm.

ALGORITHM:

- 1.Start the process
- 2.Declare the size
- 3.Get the number of pages to be inserted
- 4.Get the value
- 5.Declare counter and stack
- 6.Select the least frequently used page by counter value
- 7.Stack them according the selection.
- 8.Display the values
- 9.Stop the process

PROGRAM:

```
#include <stdio.h>

int main() {
    int frames, pages, i, j, k, pos, max, faults = 0;

    printf("Enter number of frames: ");
    scanf("%d", &frames);

    printf("Enter number of pages: ");
    scanf("%d", &pages);

    int ref[pages], frame[frames];

    printf("Enter page reference string:\n");
    for (i = 0; i < pages; i++)
        scanf("%d", &ref[i]);

    for (i = 0; i < frames; i++)
        frame[i] = -1;

    for (i = 0; i < pages; i++) {
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```
int found = 0;

for (j = 0; j < frames; j++) {
    if (frame[j] == ref[i]) {
        found = 1;
        break;
    }
}

if (found) {
    printf("\nPage %d -> Hit", ref[i]);
} else {
    int empty = -1;
    for (j = 0; j < frames; j++) {
        if (frame[j] == -1) {
            empty = j;
            break;
        }
    }

    if (empty != -1) {
        frame[empty] = ref[i];
    } else {
        max = -1;

        for (j = 0; j < frames; j++) {
            int next = -1;

            for (k = i + 1; k < pages; k++) {
                if (frame[j] == ref[k]) {
                    next = k;
                    break;
                }
            }

            if (next == -1) { // not used again
                pos = j;
                break;
            }
        }
    }
}
```

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```
        if (next > max) {
            max = next;
            pos = j;
        }
    }

    frame[pos] = ref[i];
}

faults++;
printf("\nPage %d -> Fault", ref[i]);
}
printf("\tFrames: ");
for (j = 0; j < frames; j++) {
    if (frame[j] != -1)
        printf("%d ", frame[j]);
    else
        printf("- ");
}

printf("\n\nTotal Page Faults = %d\n", faults);

return 0;
}
```

OUTPUT:

```
Enter number of frames: 3
Enter number of pages: 7
Enter page reference string:
7 0 1 2 0 3 0
Page 7 -> Fault Frames: 7 - -
Page 0 -> Fault Frames: 7 0 -
Page 1 -> Fault Frames: 7 0 1
Page 2 -> Fault Frames: 2 0 1
Page 0 -> Hit Frames: 2 0 1
Page 3 -> Fault Frames: 2 0 3
Page 0 -> Hit Frames: 2 0 3
```

Total Page Faults = 5